Critical Coverage and the Saturation Depth Problem

Reducing the BFS Window-Depth Asymptotics to a Single Coverage Observable

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Abstract

The auto-calibrated saturation depth $n_1(q)$ of the breadth-first exploration of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ was the central open asymptotic variable of O25–O28 [4, 1]. This note makes two points. First, the depth problem is not a vector-space filling problem: at the window edge the cumulative Weil–fingerprint rank is far from full, so n_1 is the depth at which the creation of new directions per explored vertex drops below a fixed threshold — a stop of marginal novelty. Second, and consequently, the entire asymptotic content of n_1 collapses onto a single dimensionless observable, the *critical coverage*

$$x_1(q) = \frac{|B_n|_{n=n_1}}{q^2}, \qquad \text{through the exact identity} \qquad n_1(q) = \left(\frac{x_1(q)}{C_{\mathrm{Heis}}}\right)^{1/4} \sqrt{q},$$

where $|B_n| \sim C_{\mathrm{Heis}}\, n^4$ (homogeneous dimension $D = 4$, $C_{\mathrm{Heis}} \approx 0.427$) is the Heisenberg ball growth. The reduction rests on two robust facts — $|B_n| \sim C_{\mathrm{Heis}}\, n^4$ and the diffusive spectral gap $\lambda_2 \approx 4\pi^2/q^2$ — and one scaling ansatz. Empirically x_1 falls from 0.61 at $q = 101$ to 0.15 at $q = 601$, which excludes the constant-coverage (clean $\sqrt{q}$) scenario and establishes $n_1(q)/q \to 0$. We isolate the governing tension $\Phi'(x_1) = \varepsilon_{\mathrm{sat}}\, q$ and state the analytical programme: derive the asymptotic law of $x_1(q)$ from a shellwise Weil–BFS novelty estimate controlled by λ_2, of which the depth law $n_1(q)$ is then a corollary.

Keywords. critical coverage; BFS saturation depth; window-depth asymptotics; marginal novelty; Weil representation; Heisenberg group; spectral gap; projective capacity; non-injective projection; spectral admissibility.

Contents

1 The saturation-depth problem

Let q be an odd prime, $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and $\mathrm{Cay}(G_q, S_q)$ its Cayley graph with symmetric generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. Breadth-first search from the identity produces spheres S_n and balls $B_n = \bigcup_{m\le n} S_m$. Each shell carries Weil $k = 3$ fingerprint vectors in $\mathbb{C}^q$; their cumulative Gram–Schmidt rank $r(n) = \sum_{m\le n} \Delta r(m)$ is monotone and bounded by q, and the capacity-novelty density is

$$\sigma(n) = \frac{\Delta r(n)}{|S_n|}. \tag{1}$$

The auto-calibrated fitting window of O12 [2] has upper edge

$$n_1(q) = \text{last } n \text{ with } \bar{\sigma}(n) > \varepsilon_{\mathrm{sat}}, \qquad \varepsilon_{\mathrm{sat}} = 10^{-3}, \tag{2}$$

where $\bar{\sigma}$ is the mean over the conjugate-pair probe blocks of (1).

Remark 1 (The depth is a marginal-novelty stop, not a filling time)**.** At the window edge the cumulative rank is far from full: $r(n_1) \approx 137/307$ at $q = 307$ and $183/601$ at $q = 601$. So n_1 is *not* the depth at which the fingerprints span $\mathbb{C}^q$. By (2) it is the depth at which the marginal rank gained per explored vertex, $\sigma(n)$, first falls below the fixed threshold $\varepsilon_{\mathrm{sat}}$. This reformulation is model-free and survives independently of the eventual asymptotic law; it is also the form that transfers to the per-generation depths n_g, which are the same kind of object (a marginal-novelty stop read off the relaxation trajectory, not a rank count).

O25–O28 [4, 1] left the asymptotic law of $n_1(q)$ as the central open direction. O28 settled its gross shape model-free: the linear hypothesis $n_1 \sim \hat{\alpha}\, q$ calibrated on $q \le 211$ fails out of sample (it overpredicts n_1 by up to +88% at $q = 601$), and $n_1(q)/q$ decreases monotonically to 0.032, so $n_1(q)/q \to 0$. The asymptotic law itself remained open. The present note reduces that open law to a single observable.

2 The critical coverage and the reduction

Two ingredients of the depth law are already understood.

Proposition 1 (Volume growth, $D = 4$)**.** On $\mathrm{Cay}(G_q, S_q)$ the homogeneous dimension is $D = 4$ (Bass–Guivarc'h; O9 [3]), so for $n \ll q$,

$$|B_n| = C_{\mathrm{Heis}}\, n^4\, (1 + o(1)), \qquad C_{\mathrm{Heis}} \approx 0.427, \tag{3}$$

with shell ratio $|S_n|/n^3 \to 4C_{\mathrm{Heis}} \approx 1.71$. The fitted exponent is $p = 4.003$.

Proposition 2 (Diffusive spectral gap)**.** The algebraic connectivity of $\mathrm{Cay}(G_q, S_q)$ satisfies

$$\lambda_2(L_{G_q}) \approx \frac{4\pi^2}{q^2}, \tag{4}$$

the gap of the horizontal abelianisation torus $(\mathbb{Z}/q\mathbb{Z})^2$ (numerically $\lambda_2\, q^2 \to\approx 39 \to 4\pi^2 = 39.48$ over $q \in \{5, \ldots, 19\}$). The walk relaxes diffusively, and its horizontal phase space has q^2 vertices.

These fix the natural coverage variable: a diffusive walk with gap q^{-2} on a space of horizontal extent q^2 is organised by the reduced volume $x = |B_n|/q^2$. Define the *critical coverage* at the window edge:

Definition 1 (Critical coverage)**.**

$$x_1(q) = \frac{|B_n|_{n=n_1}}{q^2} = \frac{C_{\mathrm{Heis}}\, n_1(q)^4}{q^2}\, (1 + o(1)). \tag{5}$$

Inverting (3) at the crossing volume $|B_n|_{n=n_1} = x_1(q)\, q^2$ gives the organising identity of this note.

Proposition 3 (Reduction of the depth problem to the coverage problem)**.** Exactly (to leading order in (3)),

$$n_1(q) = \left(\frac{x_1(q)}{C_{\mathrm{Heis}}}\right)^{1/4} \sqrt{q}. \tag{6}$$

Hence the asymptotic law of $n_1(q)$ *is* the asymptotic law of $x_1(q)$: a constant x_1 gives $n_1 = \Theta(\sqrt{q})$; a decaying x_1 gives a sub-$\sqrt{q}$ law. In every case $x_1 \le O(1)$ keeps $n_1 = O(\sqrt{q})$, so $n_1/q \to 0$.

The protagonist of the saturation-depth problem is therefore $x_1(q)$, not $n_1(q)$.

3 Empirical law of the critical coverage

Table 1 reports $n_1(q)$ (auto-calibrated as in (2), reproducing the O28 production values exactly on $q \le 211$) and the derived coverage $x_1(q)$ via (5).

Table 1: Auto-calibrated saturation depth $n_1(q)$, the ratio n_1/q, and the critical coverage $x_1(q) = C_{\mathrm{Heis}}\, n_1^4/q^2$ with $C_{\mathrm{Heis}} = 0.427$. Values $q \le 211$ reproduce O28 [1]; $q \ge 307$ are out of sample.

q	$n_1(q)$	$n_1(q)/q$	$x_1(q)$
29	4	0.138	0.130
61	8	0.131	0.470
101	11	0.109	0.613
151	13	0.086	0.535
211	14	0.066	0.368
307	16	0.052	0.297
401	17	0.042	0.222
503	19	0.038	0.220
601	19	0.032	0.154

Two readings of the same table. The depth ratio n_1/q falls monotonically $0.138 \to 0.032$: the linear law is dead and $n_1/q \to 0$ (O28). The coverage x_1 falls fast, $0.61 \to 0.15$ over $q = 101 \to 601$, tracking $x_1 \sim q^{-0.76}$ on the available range, with a brief integer-rounding stall at $q = 401$–503 (where n_1 sits on the integer 19) before resuming. The striking empirical fact is this sustained fall of the coverage: the vertex coverage at which the crossing $\sigma(n_1) = \varepsilon_{\mathrm{sat}}$ occurs keeps moving to ever-lower values as q grows.

4 Exclusion of the constant-coverage scenario

By Proposition 3 the candidate depth laws are candidate coverage laws:

- $x_1 \to \mathrm{const} \implies$ clean $\Theta(\sqrt{q})$;
- $x_1 \propto (\ln q)^{-4} \implies n_1 \approx \kappa\sqrt{q}/\ln q$;
- $x_1 \propto q^{-2/3} \implies n_1 \propto q^{1/3}$.

The point $q = 601$ excludes the constant-coverage scenario. The apparent plateau $x_1 = 0.222\,(401) \to 0.220\,(503)$ suggested convergence, which would have revived a clean asymptotic $\sqrt{q}$ with $\kappa \approx 0.85$; but the constant-x_1 reading predicts $n_1(601) = 21$, whereas the measurement gives 19. The stall was an integer-rounding artefact (n_1 held the value 19 across $q = 503$ and the $q = 601$ neighbourhood), not a limit: x_1 resumed its decrease to 0.154.

The two surviving readings, $n_1 \approx \kappa\sqrt{q}/\ln q$ and $n_1 \propto q^{1/3}$, are both sub-$\sqrt{q}$ and remain numerically indistinguishable on the present data (each predicts $n_1(601) = 20$ against the measured 19; they differ by less than one unit in n_1 out to $q \sim 10^3$). The effective exponent observed over $101 \le q \le 601$ is ≈ 0.30–0.33, below $\frac{1}{2}$, but this is a range observation, not an established asymptotic value: the n_1 are small integers and the $503 \to 601$ step is $19 \to 19$, which dominates any power fit. What is established for n_1 is therefore: sub-linear, incompatible with $\Theta(q)$, and — over the observed range — below $\Theta(\sqrt{q})$; the precise asymptotic law is the open law of $x_1(q)$.

5 The governing tension

The reduction (6) is exact; what is *not* yet proved is the law of x_1. The structural route to it is the coverage-scaling ansatz suggested by Propositions 1–2:

$$r_q(n) \approx q\,\Phi\left(\frac{|B_n|}{q^2}\right), \qquad \Phi \text{ smooth, } \Phi(0) = 0, \ \Phi \text{ increasing.} \tag{$\star$}$$

This is a scaling collapse: the q-dependence of the whole rank trajectory is assumed to enter only through the reduced coverage $x = |B_n|/q^2$. Granting ($\star$), and using $\sigma(n) = \Delta r/|S_n| \approx \mathrm{d}r/\mathrm{d}|B_n|$ (the marginal density of Remark 1), one gets $\sigma(n) = \Phi'(x)/q$, so the window criterion $\sigma(n_1) = \varepsilon_{\mathrm{sat}}$ reads

$$\Phi'(x_1) = \varepsilon_{\mathrm{sat}}\,q. \tag{7}$$

The right-hand side *grows with* q while $\varepsilon_{\mathrm{sat}} = 10^{-3}$ is fixed. Equation (7) is the single mathematical front of the problem: a fixed threshold against the explicit $1/q$ in σ. It forces the crossing coverage x_1 into the steep part of Φ as q grows, hence the observed decay of x_1; whether that decay is logarithmic or a residual power is exactly the unresolved behaviour of $x_1(q)$, and explaining it — rather than fitting an exponent — is the task.

6 Analytical programme for the law of $x_1(q)$

The target is a structural law for $x_1(q)$, from which $n_1(q)$ follows by (6). The proposed route has three steps.

1. **Shellwise novelty estimate.** Bound the per-shell rank increment $\Delta r(n)$ of the Weil $k = 3$ fingerprints by the mixing of the BFS distribution at depth n, controlled by the diffusive gap $\lambda_2(L_{G_q})$ of (4). This is the missing derivation of the ansatz ($\star$): it should produce Φ, not assume it.
2. **Crossing analysis.** Solve (7) with the derived Φ to obtain $x_1(q)$, and hence the asymptotic law and any logarithmic or power correction.
3. **Falsifiable prediction.** Extend the campaign to $q \in \{809, 1009, 1201\}$ to test the predicted $x_1(q)$ directly (the coverage, not a fitted exponent, is the observable).

The same machinery is the bridge to the per-generation depths n_g: by Remark 1 they are marginal-novelty stops governed by the same relaxation λ_2, so a law for x_1 is the template for the corresponding per-generation coverages.

7 Status

- *Proved / robust:* $D = 4$ growth, $|B_n| \sim C_{\mathrm{Heis}} n^4$ (Prop. 1); $\lambda_2 = \Theta(1/q^2)$, $\approx 4\pi^2/q^2$ (Prop. 2); the exact reduction identity (6) (Prop. 3); $n_1(q)/q \to 0$ (model-free, O28).
- *Established empirically (range-limited):* $x_1(q)$ decreases ($0.61 \to 0.15$ over $q = 101$–601); the constant-coverage scenario is excluded; the observed effective exponent of n_1 is $\approx$ 0.30–0.33.
- *Open:* the asymptotic law of $x_1(q)$ (logarithmic vs. power), the ansatz ($\star$) and its Φ, and the resolution of the tension (7).

Data and Code Availability. The window-depth measurement (`n1_resumable.py`, multi-core `n1_parallel.py`), the ball-growth constant (`cheis_ballgrowth.py`, $C_{\mathrm{Heis}} \approx 0.427$, $D = 4$), and the spectral-gap scaling (`lambda2_scaling.py`, $\lambda_2 q^2 \to 4\pi^2$) are in the `simulation/spectral/o25` directory of the Cosmochrony repository. Both depth scripts reproduce the O28 values $\{29, 61, 101, 151, 211\} \mapsto \{4, 8, 11, 13, 14\}$ exactly.

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